Representational Incompleteness: No Self-Modeling System Can Capture Its Own Diagonal

A Machine-Checked Barrier with Six Closed Escape Routes

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Abstract

Summit theorem (`representational_incompleteness`). Any system that models itself—any $s : A \to A \to B$ whatsoever—is *necessarily incomplete* with respect to its own diagonal behavior: the function $a \mapsto f(s\,a\,a)$ is never equal to any row $s\,a_0$ of the self-model, for every fixed-point-free $f : B \to B$ and every $a_0 \in A$. No surjectivity, totality, or cardinality assumption on s is required. If s is claimed to be total/surjective, it simply cannot exist.

This is **representational incompleteness**: the analogue of Gödel incompleteness, not for formal theories and provability, but for any system that models itself via a parametric family of functions. The blind spot is structural—not a resource limitation—and persists for every codomain admitting a fixed-point-free endomorphism (including $\mathbb{N}$ with successor and Bool with negation). Unlike Gödel incompleteness, representational incompleteness requires no formal theory, no arithmetic, and no coding; it holds for every parametric self-model regardless of computability, cardinality, or totality.

Topological reinforcement. A second, independent strand proves that the barrier is not merely diagonal but *topological*: a general lemma states that any two spaces whose intrinsic planar-chart boundaries are subtype-incompatible are not homeomorphic. The obstacle is **wrong type**, not **inexhaustible queue**—a finite-data invariant that no amount of computation, scaling, or architectural extension can change. The lemma applies to an *open-ended class* of target spaces (Möbius strip, Klein bottle, projective plane, or any non-orientable surface with incompatible boundary); the closed cylinder and Möbius strip are the minimal fully discharged compact witness.

No escape. Six adversarial retreat routes are formally closed: infinite regress, finite-order iterate collapse, cross-object readout (both sides), partial self-models, decidable self-reference, and model substitution. Each is a proved theorem in Lean 4, checked against Mathlib, with zero `sorry` and no axioms beyond Lean 4's standard foundation (propositional extensionality, quotients, choice).

Three pillars. The result rests on three mutually reinforcing pillars: (1) *diagonal exclusion*—every self-model has a blind spot shaped like its own diagonal; (2) *topological obstruction*—a general lemma defining an open-ended class of space-pairs that are not homeomorphic, detected by a finite-data boundary invariant impervious to computation; (3) *ontological sort wall*—the `OntologicalSlot` construction enforces that the self-representation morphism and any global element remain definitionally disjoint at every iterate depth; the disjointness is proved robust under arbitrary injective relabeling (`ClosureProperties`), so it is a structural invariant, not a notational artifact.

Closure properties. The diagonal exclusion is proved robust under composition, reindexing, post-composition, embedding, product formation, multi-agent delegation, and nested simulation.

A formal *bridge hypothesis* P3 makes the modeling claim "the system has a parametric self-model" explicit and conditional, parallel to P2 for consciousness. A `TopologicalSelfModelingSystem` structure formally bridges Pillars 1 and 2.

Consequence. In every neighboring debate—consciousness, symbolic AI, semantic grounding, measurement chains, LLM scaling, AGI self-improvement, and the simulation hypothesis—the six closed escape routes leave every remaining admissible path requiring *structural change* to the topological type of the substrate, not quantitative extension within a fixed type. Scaling, iterating, partial modeling, cross-object readout, finite-depth loops, and decidable restriction are all formally inadmissible.

What fills the blind spot. The blind spot is located at the self-referential diagonal. A functioning self-modeling system *does* produce diagonal behavior, but its self-model *cannot* represent that behavior. Therefore the substrate—the underlying machinery executing the system—is the **formally necessary complement** to any parametric self-model. No self-improvement, delegation, or scaling can substitute for it. Self-awareness with perfect self-transparency is the one thing that is structurally impossible; self-awareness with a diagonal opacity is what the formalism predicts and what empirical evidence confirms.

1 Introduction

1.1 The Problem and the Summit Theorem

When a system models itself—by iterating a fixed construction, by encoding its own states, by interpreting its own symbols—a familiar worry arises: infinite regress, or diagonal limitations, or both. But those concerns are usually framed *computationally*: the regress does not terminate, the self-interpreter cannot be total, the Gödel sentence is undecidable. The implicit hope is that with enough resources—more compute, richer architectures, deeper self-models—the obstacle might shrink or be routed around.

This paper proves it cannot. The headline result is:

Theorem 1.1 ([Novel] Representational Incompleteness)**.**
(`representational_incompleteness`)
Let B be a type with a fixed-point-free endomorphism $f : B \to B$. For ***any*** *$s : A \to A \to B$ (no surjectivity, totality, or cardinality assumption):*

(a) ***Diagonal exclusion:*** *for every $a_0 \in A$, $s\, a_0 \neq \lambda a.\, f(s\, a\, a)$.*

(b) ***Total coverage impossible:*** $\neg(\forall g : A \to B,\ \exists a,\ s\, a = g)$.

Every self-modeling system is *necessarily incomplete*: its own diagonal behavior is precisely the information it cannot represent. If s is claimed to be total and surjective, it simply cannot exist. This is **representational incompleteness**—the analogue of Gödel incompleteness, not for formal theories and provability, but for any system that models itself via a parametric family of functions. The blind spot is structural, not a resource limitation, and persists for $B = \mathbb{N}$ (successor), $B = \mathrm{Bool}$ (negation), and every other codomain admitting a fixed-point-free endomorphism.

A functioning self-modeling system *does* produce diagonal behavior—it processes self-referential inputs and generates outputs. Since no row of its own self-model matches that behavior, the diagonal behavior must be generated by the underlying *substrate*: the physical or computational machinery executing the system. The substrate is the formally necessary complement to any parametric self-model (§7.9.3).

A second, independent strand reinforces this with a **topological** barrier: a general lemma showing that *any* two spaces whose intrinsic planar-chart boundaries are subtype-incompatible are

not homeomorphic—an open-ended class of obstructions that no amount of iteration, scaling, or continuous deformation within a fixed topological type can remove. And we close, one by one, every standard adversarial route by which one might hope to circumvent either barrier.

The technical development is in Lean 4, checked against Mathlib, with zero `sorry` and no axioms beyond Lean 4's standard foundation; §7.3–§7.12 relate the results to consciousness, symbolic grounding, diagonal arguments, the simulation hypothesis, AI/AGI scaling, and observer–apparatus topologies.

This is not Gödel repackaged

Representational incompleteness differs from Gödel incompleteness in every structural respect. Gödel requires a formal theory rich enough for arithmetic, a Gödel coding, and a self-referential *sentence*; the blind spot is an *unprovable truth*. Representational incompleteness requires only a type B with a fixed-point-free endomorphism and *any* $s : A \to A \to B$; the blind spot is an *unrepresentable function*—the system's own diagonal. No formal theory, no arithmetic, no coding, no computability assumption is needed. The result is **stronger** in one respect: it holds for every s regardless of whether s is computable, whereas Gödel- and Turing-style arguments require the system to be recursively axiomatizable or computable.

1.2 Summary of Results

The core results are **representational incompleteness** (Theorem 1.1) and a **universal topological obstruction**—reinforced by a proof that the six most natural adversarial attempts to evade them all fail.

The topological obstruction. A general lemma (Theorem 4.4) states: if two spaces admit subsets characterized as exactly the non-`ChartableR2` points, and if those boundary *subspaces* are not homeomorphic, then the ambient spaces are not homeomorphic. This is a *finite-data invariant*: two spaces are either homeomorphic or not, regardless of how many modeling steps one enumerates. The lemma defines an *open-ended class* of obstructed pairs—any two spaces meeting the three hypotheses, regardless of genus, orientability, or dimension—rather than a single hard-wired example. The closed cylinder and Möbius strip serve as the **minimal-case witness** (**M-FINAL**); Klein bottles, projective planes, higher-genus non-orientable surfaces, etc., would equally suffice. Any future domain whose state space matches the same boundary pattern imports the same conclusion with no new proof.

Why this is not "just" infinite regress—or "just" Gödel. The categorical strand yields infinitely many distinct meta-levels, but infinity is not the deepest point. The topological obstruction is independent: it says the barrier is **wrong type**, not **inexhaustible queue**. Nor is this "just Gödel": representational incompleteness requires no formal theory, no arithmetic, no coding, and no computability assumption on s. The blind spot is a *function*, not a sentence, and no theory extension can close it. Section 7.10 provides a detailed comparison on six independent structural axes.

The topological barrier is independent of diagonalization

The topological strand (Theorem 4.4) is *completely orthogonal* to Cantor, Gödel, Lawvere, and every other diagonal argument. It detects a *finite-data invariant*: the connectivity pattern of the non-chartable boundary. No amount of iteration, scaling, architectural reparameterization, or continuous deformation within a fixed topological type can alter this invariant. The obstacle is **wrong type**, not **inexhaustible queue**—and the general lemma defines an *open-ended class* of obstructed pairs, not a single hard-wired example.

No escape. Six adversarial retreat routes—infinite regress, finite-order collapse, cross-object readout, partial self-models, decidable self-reference, and model substitution—are each formally closed by a proved theorem (§7.11). Together they leave an adversary with a clean dichotomy: either an infinite injective regress with the Lawvere wall intact, or a finite loop too shallow for full self-capture—and in either case the topological barrier stands.

Bridges and conditional readings. Cross-domain structures (`SymbolSystem`, `LawvereRegressBridge`, bundled `ChartableR2BoundaryModel` instances, `QuantumObserverChainHypothesis`) make the obstruction's *hypothesis shape* explicit for grounding, diagonal, consciousness (under premise P2), and observer–readout readings. Because the topological obstruction is a general lemma over a class of space-pairs, each bridge inherits the full generality automatically. An extended master theorem adjoins all three layers (diagonal, topological, categorical) in one checkable bundle.

Utility. For any application the workflow is: (1) encode the system's state space in sufficient topological detail to verify the boundary biconditionals, (2) compare boundary subspaces up to homeomorphism, (3) invoke the general lemma or a bundled model. If the premises hold, the obstruction applies; if they fail, the obstacle was never licensed. That conditional structure is what keeps the mathematics honest while delivering a *portable* gate.

1.3 Contributions and Novelty Claims

Individual ingredients are classical: Lawvere's diagonal (1969), textbook topology of surfaces, Cantorian regress patterns. The contribution is a **unified, adversarially checked, machine-verified** landscape in which representational incompleteness, topological obstruction, and six closed escape routes coexist in one artifact, so that "compute harder" is never confusable with "change homeomorphism type." Specific novel elements:

(1) `OntologicalSlot` — inductive tagging of morphism vs. global-element slots; makes the Lawvere wall explicit inside the type discipline. [[Novel]]

(2) `ChartableR2` **spine + seam–equator charts** — intrinsic $\mathbb{R}^2$ chartability as a definable boundary sensor; constructive charts through the Möbius quotient seam including the equator via piecewise unfolding. The Möbius strip serves as the minimal proof witness; the predicate applies uniformly to all surfaces. [[Novel] technique]

(3) **General boundary-contrast obstruction** — a single reusable, machine-checked lemma defining an *open-ended class* of obstructed pairs: any two spaces with incompatible `ChartableR2`-co-characterized boundaries are not homeomorphic—regardless of genus, orientability, or dimension. To our knowledge, *not* standard in the philosophy-of-mind or AI-safety literatures. [[Novel]]

(4) **Cross-domain bridge structures** — `SymbolSystem`, `LawvereRegressBridge`, bundled `ChartableR2BoundaryModel` data, `QuantumObserverChainHypothesis`: these make the obstruction's hypothesis shape explicit per domain. [[Novel]]

(5) **No-escape horn closures** — six formally blocked adversarial escape routes (iterate collision, section–retraction on both sides, partial-model diagonal exclusion, decidable Lawvere, wall persistence at every depth). The integrated closure package is, to our knowledge, new. [[Novel]]

(6) **Closure properties and bridge hypothesis P3** — compositional inheritance of the diagonal exclusion through post-composition, reindexing, embedding, and product formation; type-level cross-object readout; a formal `ComputationalSelfModel` bridge (P3) making the AI/computational modeling claim explicit and conditional; slot robustness theorems proving the endo/obj wall survives any injective relabeling; per-pillar full-closure witnesses; and a `TopologicalSelfModelingSystem` structure formally bridging Pillars 1 and 2. [[Novel]]

1.4 Paper Organization

The paper is structured around three layers that build toward a single conclusion.

Layer 1: Diagonal & categorical (§2)—representational systems, regress, Lawvere wall, `OntologicalSlot`, iterate dichotomy, core master theorem.

Layer 2: Topological obstruction (§§3–6)—minimal-case witnesses (cylinder, Möbius strip), `ChartableR2` boundary sensor, general obstruction lemma, seam charts, M-FINAL.

Layer 3: Integration & closure (§7)—extended master theorem, instantiations (consciousness, grounding, diagonal arguments, QM, simulation hypothesis, LLM scaling, AI/AGI), anatomy of the blind spot (§7.9), horn-by-horn no-escape analysis (§7.11), admissibility (§7.12).

§§8–10: limitations, related work, conclusion. Appendices: Lean module map, key theorem index, reproduction.

2 The Representational Regress

This section builds the categorical machinery that feeds into the summit theorem: the endomorphism tower, the Lawvere wall, and the `OntologicalSlot` distinction. Together they establish that self-representation via iterated endomorphisms either produces an infinite regress with the endo/obj wall intact, or collapses to a finite loop too shallow for injective self-capture—setting the stage for the diagonal exclusion of Theorem 1.1.

2.1 The `RepresentationalSystem` Structure

Definition 2.1 (`Basic.lean`: `RepresentationalSystem`)**.**
A representational system *is a tuple* $(C, A, \mathrm{represent}, \mathrm{iter_injective})$ *where:*

- C *is a category,*
- A *is an object of* C*,*
- $\mathrm{represent} : A \to A$ *is an endomorphism,*
- $\mathrm{iter_injective}$*: for all* $n \neq m$*,* $\mathrm{represent}^n \neq \mathrm{represent}^m$ *in* $\mathrm{End}(A)$*.*

2.2 Non-Termination and Unboundedness

Theorem 2.2 ([Known math] `Regress.lean`). *For any representational system R:*

(a) (`regress_no_termination`) Distinct indices $n \neq m$ yield distinct meta-level arrows.

(b) (`regress_iterates_unbounded`) For every bound B, there exists a level $L > B$ with a distinct iterate.

(c) (`regress_is_infinite`, `meta_range_infinite`) The set of meta-levels is infinite.

Remark 2.3 (What if `iter_injective` fails?). *The hypothesis* iter_injective *is* bundled*, not derivable from a bare endomorphism. An adversary might weaken it: suppose* $\text{represent}^n = \text{represent}^m$ *for some $n \neq m$. `NoEscapeClosure` formalizes the consequence: let $e :=$* End.of(represent)*; if $e^{n+d} = e^n$ with $d > 0$ then every power e^k equals some e^i with $i < n + d$ (`IterateClosure.End.exists_lt_of_pow_collision`). `IterateClosure.End.pow_iterate_dichotomy` records the **dichotomy**: for every monoid element e, either (i) distinct natural-number powers are all distinct (an infinite injective $\mathbb{N}$-tower), or (ii) the set of* distinct *powers is bounded by a finite index. In case (ii) the system has only finitely many representational layers—it loops. Either route blocks "just enough" self-representation: one gives an infinite regress; the other gives a finite loop too impoverished to host injective iterate depth.*

2.3 The Lawvere Wall

Theorem 2.4 ([Known math] `Lawvere.lean`). *(`lawvere_fixed_point_Type`) If $s : A \to (A \to B)$ is surjective, then every $f : B \to B$ has a fixed point.*

Theorem 2.5 ([Known math] `LawvereCCCType.lean`). *(`lawvere_fixed_point_MonoidalClosedType`) The same conclusion holds when the universal-enumerator hypothesis is recast as surjectivity of* curry(lawvereBinary s) *in* MonoidalClosed(Type u)*.*

2.4 The `OntologicalSlot` Construction

Definition 2.6 ([Novel] `Basic.lean`: `OntologicalSlot`). *For a category C with terminal object and an object A:*

$$\text{OntologicalSlot}(C, A) \;:=\; \text{endo}(f : A \to A) \;\mid\; \text{obj}(g : \top_C \to A)$$

an inductive type with two constructors, disjoint by definition.

Theorem 2.7 ([Novel] `FixedPoints.lean`, `LawvereRegressBridge.lean`).

(a) (`lawvere_wall_is_not_dissolution`) For any representational system R and any global element $\text{fp} : \top_C \to A$, OntologicalSlot.endo(R.represent) $\neq$ OntologicalSlot.obj(fp).

*(b) (`lawvere_fixed_point_stays_representational`) The wall holds at **every iterate depth**: for all $n : \mathbb{N}$ and all $c : C$,* OntologicalSlot.endo(represent^n) $\neq$ OntologicalSlot.obj(c).

2.5 Why `OntologicalSlot` Is Novel: The Wall That Lawvere Does Not Build

Lawvere's fixed-point theorem (1969) proves that if $s : A \to (A \to B)$ is surjective, then every $f : B \to B$ has a fixed point. This is a theorem about surjectivity and fixed points. It does *not* tag the fixed point as belonging to a different ontological sort from the morphism that produced it.

In standard category theory, the endomorphism monoid $\mathrm{End}(A)$ and the set of global elements $\mathrm{Hom}(\mathbf{1}, A)$ are both sets of morphisms in the ambient category. There is no formal marker distinguishing "this morphism is a process acting on A" from "this morphism is a state of A selected from the terminal object." The two live in different hom-sets, but no inductive tag enforces their disjointness inside a single data structure.

`OntologicalSlot` (Definition 2.6) fills this gap. It is an inductive type with two constructors—$\mathrm{endo}(f)$ and $\mathrm{obj}(g)$—that are *definitionally disjoint* by constructor discrimination. The wall theorem (Theorem 2.7) then proves that for any representational system, the self-representation morphism represent (tagged as endo) is never equal to any global element fp (tagged as obj), regardless of how many times represent is iterated. The Lawvere fixed point exists as an object-witness; it does *not* dissolve the endo/obj wall.

Philosophical significance. In philosophy of mind, the intuition that "the *process* of self-representation and the *product* of self-representation inhabit different ontological categories" is widespread but usually left informal. `OntologicalSlot` is its machine-checked formalization. The earlier "Transputation" argument (Spivack, 2025) claimed that in the base case of awareness aware of itself ($A \to A$), the roles of knower, known, and knowing "collapse"—the process of being aware and the state of being aware cannot be distinguished phenomenologically. `OntologicalSlot` formalizes the *impossibility* of that collapse within any categorical representation: the iterate $\mathrm{represent}^n$ (a morphism, an endo) can never equal a global element $\mathrm{fp} : \mathbf{1} \to A$ (a state, an obj). If consciousness involves a genuine identification of process and product, then no categorical self-model can achieve it.

The wall that Lawvere does not build

Lawvere's theorem guarantees a fixed point. `OntologicalSlot` guarantees that the fixed point remains an *object*-witness, never collapsing into the *morphism*-process that generated it. The "wall" between endo and obj is a definitional invariant that persists at every iterate depth—not proved by Lawvere, not present in standard category theory, and not available in any prior formalization of self-reference.

Robustness: the wall is structural, not notational. An adversary might object: "The endo/obj disjointness is trivially true by constructor discrimination—it is a notational convention, not a mathematical barrier." This objection is addressed by the *robustness theorems* in `Closure Properties.lean`:

- `OntologicalSlot.wall_survives_injection`: for any injective relabeling $\varphi : \mathrm{OntologicalSlot} \to T$, the wall persists: $\varphi(\mathrm{endo}(f)) \neq \varphi(\mathrm{obj}(c))$.
- `OntologicalSlot.wall_collapse_requires_noninjective`: the *only* way to collapse endo and obj into the same image is via a non-injective map—one that deliberately identifies distinct things.
- `OntologicalSlot.iterate_tower_injective`: distinct iteration depths $n \neq m$ yield distinct endo-slots, so the iterate tower is faithfully represented in slot space.

The wall is therefore a *structural invariant of the type discipline*: it survives any relabeling that respects distinctness. To erase it, one must use a function that conflates distinct objects by design—which is not an escape but a concession.

2.6 The Core Master Theorem

Theorem 2.8 (`Main.lean`). *(`representational_regress_master`) For any representational system R:*

$$\text{representational_regress_master_claim}(R)$$

i.e., the conjunction of unbounded regress, Lawvere wall non-dissolution, and T_2 invariance under homeomorphism.

3 Topological Models: Minimal-Case Witnesses

The general obstruction (Theorem 4.4) applies to any pair of spaces with incompatible `ChartableR2`-detected boundaries. This section establishes the closed cylinder and Möbius strip as the **simplest fully discharged witness pair**: two compact surfaces whose boundary subtypes are topologically incompatible.

3.1 The Closed Cylinder

Definition 3.1 (`CylinderBoundary.lean`). ClosedCylinder $:= S^1 \times [0,1]$ *with the product topology and manifold structure* $\mathfrak{R}_1 \times \mathfrak{R}_1^{\partial}$.

Theorem 3.2 ([Known math] `CylinderBoundary.lean`). *The manifold boundary of* ClosedCylinder *decomposes as:*

$$\partial(\text{ClosedCylinder}) = F_{\text{bot}} \sqcup F_{\text{top}}$$

where $F_{\text{bot}} = S^1 \times \{0\}$ and $F_{\text{top}} = S^1 \times \{1\}$ are each connected, and their union is disconnected.

3.2 The Möbius Strip

Definition 3.3 (`CylinderMobius.lean`). *The* Möbius equivalence $\sim$ *on* $[0,1]^2$ *is the equivalence relation generated by* $(0,t) \sim (1, 1-t)$ *for* $t \in [0,1]$, *extended by reflexivity, symmetry, and transitivity. The* Möbius strip *is* MobiusStrip $:= [0,1]^2/{\sim}$.

Theorem 3.4 (`CylinderMobius.lean`).

(a) *(`mobiusRelOGraph_isClosed`) The graph of $\sim$ is closed in $[0,1]^2 \times [0,1]^2$.*

(b) *(`CompactSpace MobiusStrip`)* MobiusStrip *is compact (quotient of compact).*

(c) *(`instT2SpaceMobiusStrip`)* MobiusStrip *is Hausdorff (closed graph + compact domain).*

(d) *(`isConnected_mobiusStripBoundary`) The boundary band $\partial_{\text{model}} := \pi(\{t = 0\} \cup \{t = 1\})$ is connected (one component).*

3.3 Boundary Subspace Contrast

Theorem 3.5 (`MobiusCylinderBoundaryContrast.lean`).
(`mobiusStripBoundary_not_homeomorphic_closedCylinderBoundaryUnion`)

$$\text{IsEmpty}(\text{Subtype}(\partial_{\text{Mob}}) \simeq_t \text{Subtype}(\partial_{\text{Cyl}}))\,.$$

The Möbius boundary band (connected) and the cylinder boundary union (disconnected) are not homeomorphic as subspaces.

4 The `ChartableR2` Framework

The previous sections established diagonal and categorical barriers. This section introduces the topological layer: a boundary sensor based on intrinsic $\mathbb{R}^2$-chartability, together with the general obstruction lemma (Theorem 4.4) that will deliver the class-level topological barrier.

4.1 Definition and Basic Properties

Definition 4.1 (`ChartableR2Neighbor.lean`). *A point x of a topological space X is* ChartableR2 *if there exists an open set $U \ni x$ such that $U \cong_t \mathbb{R}^2$.*

Theorem 4.2 (`ChartableR2Neighbor.lean`).

(a) (`isOpen_setOf_chartableR2`) The set of `ChartableR2` points is open.

(b) (`isClosed_setOf_not_chartableR2`) Its complement is closed.

(c) (`not_chartableR2_of_tendsto_seq_atTop`) Sequential stability: limits of non-chartable points are non-chartable.

4.2 Invariance Under Homeomorphism

Theorem 4.3 (`ChartableR2Bridge.lean`).
(`Homeomorph.chartableR2_iff`)
If $h : X \simeq_t Y$ is a homeomorphism, then $\mathrm{ChartableR2}(x) \iff \mathrm{ChartableR2}(h(x))$.

4.3 The Generic Boundary-Contrast Obstruction

Theorem 4.4 ([Novel] `ChartableR2Bridge.lean`).
(`not_homeomorphic_of_chartableR2_boundary_contrast`)
Let X and Y be topological spaces with predicates $b_X \subseteq X$ and $b_Y \subseteq Y$ such that:

(i) $\forall x,\ x \in b_X \iff \neg\,\mathrm{ChartableR2}(x)$,

(ii) $\forall y,\ y \in b_Y \iff \neg\,\mathrm{ChartableR2}(y)$,

(iii) $\mathrm{IsEmpty}(\mathrm{Subtype}(b_X) \simeq_t \mathrm{Subtype}(b_Y))$.

Then $\mathrm{IsEmpty}(X \simeq_t Y)$.

Proof sketch. Any homeomorphism $e : X \simeq_t Y$ preserves ChartableR2 (Theorem 4.3), hence maps b_X to b_Y pointwise. By e.subtype, we get a homeomorphism $\mathrm{Subtype}(b_X) \simeq_t \mathrm{Subtype}(b_Y)$, contradicting (iii). □

Theorem 4.4 subsumes the non-homeomorphism strategy of §6: any other pair of spaces satisfying the same three hypotheses is also not homeomorphic.

4.4 The Topological Barrier in Detail

Theorem 4.4 deserves sustained attention because it establishes a barrier of a fundamentally different character from anything in the Cantor–Gödel–Lawvere tradition. This subsection unpacks what the barrier is, why it is impervious to computation, and what its generality implies.

What `ChartableR2` detects. A point x of a topological space X is ChartableR2 if it has an open neighborhood homeomorphic to $\mathbb{R}^2$ (Definition 4.1). This is a *local, intrinsic* property: it depends only on the topology of X near x, not on any embedding, metric, or coordinate system. The set of ChartableR2 points is open (Theorem 4.2a); its complement—the "boundary" in the ChartableR2 sense—is closed and sequentially stable.

The key observation is that whether a point is ChartableR2 is a **finite-data invariant**: it depends on the existence of a single open neighborhood and a single homeomorphism to $\mathbb{R}^2$. Two spaces are either homeomorphic or not; this status does not change with more computation, more data, or more iterations. It is a *topological fact*, not a resource-dependent property.

Why the invariant is impervious to computation. Consider a system whose state space has a particular boundary-connectivity pattern (e.g., two disjoint connected components, like the closed cylinder). The following moves *do not change* the topological type of the state space, and therefore do not change its boundary pattern:

- **Computational scaling**—more parameters, more layers, longer context windows, more training data. These extend the system *within* a fixed topological type; they do not produce a homeomorphism to a different type.
- **Self-model iteration.** Building deeper meta-levels, stacking self-interpreters, or running represent^n for larger n. The iterate lives in the same endomorphism monoid and does not alter boundary connectivity.
- **Architectural reparameterization**—changing activation functions, attention patterns, or layer structure. If the reparameterization is continuous (a homeomorphism of the parameter space), the topological type is preserved by definition. If it is discontinuous, it is no longer "the same architecture, tuned differently"—it is a genuinely different topological space, which is precisely the kind of structural change the barrier demands.
- **Model substitution**—choosing a Klein bottle, projective plane, or higher-genus non-orientable surface as the target. The general lemma covers all of these: any two spaces with incompatible ChartableR2-co-characterized boundaries are not homeomorphic, regardless of which specific surfaces are involved.

The general lemma as an open-ended class. Theorem 4.4 is *not* a statement about the cylinder and the Möbius strip. It is a universally quantified statement: for *any* topological spaces X and Y, if (i) there exist subsets $b_X \subseteq X$ and $b_Y \subseteq Y$ that are exactly the non-ChartableR2 points, and (ii) the subspace topologies on b_X and b_Y make them non-homeomorphic, then X and Y are not homeomorphic. This defines an **open-ended class** of obstructed pairs: the class includes the cylinder/Möbius pair (the minimal witness), but it equally includes torus/Klein bottle, sphere/projective plane, and any pair of surfaces—or higher-dimensional spaces—that satisfy the three hypotheses.

Any future domain whose state space matches the boundary pattern of one side of the class inherits the obstruction with no new proof required. The adversary cannot escape by choosing a different target; the lemma is waiting for every target that has an incompatible boundary.

"Wrong type," not "inexhaustible queue." The deepest distinction between the topological barrier and diagonal/regress arguments is this: diagonal arguments say "you will always miss something—there is always one more function not in the range." The topological barrier says "you are the *wrong kind of thing*—no amount of adding, iterating, or scaling can bridge a difference in topological type." The first is an inexhaustible queue; the second is a category mismatch. Both

are formally proved, but the topological barrier is the one that cannot be routed around by "trying harder within the same framework."

The topological barrier: wrong type, not inexhaustible queue

The general boundary-contrast obstruction (Theorem 4.4) defines an **open-ended class** of topologically obstructed space-pairs. The invariant—boundary connectivity as detected by ChartableR2—is a **finite-data topological fact**: no amount of computation, scaling, iteration, or continuous deformation within a fixed topological type can alter it. The cylinder/Möbius pair is the **minimal-case witness**; the argument itself is the **general lemma**, which applies to every pair of spaces with incompatible boundaries—Klein bottles, projective planes, higher-genus surfaces, and beyond. The obstacle is not "you will always miss one more item" but "you are the wrong kind of space."

4.5 Half-Space Obstructions

Theorem 4.5 (`HalfPlaneVsPlane.lean`).
(`euclideanHalfSpace2_not_homeomorphic_euclidean2`)
IsEmpty$(H^2 \simeq_t \mathbb{R}^2)$. *Proof via punctured half-plane (star-convex $\Rightarrow$ simply connected) vs. punctured plane (not simply connected, from* exp *covering).*

Theorem 4.6 (`HalfSpaceNeighborVsPlane.lean`).
(`isEmpty_homeomorph_subtype_openH2_zero_nbhd_subtype_openPlane`)
For open $V \subseteq H^2$ with $0 \in V$ and open $O \subseteq \mathbb{R}^2$, there is no homeomorphism $V \cong_t O$.

4.6 Cylinder C4: Boundary Biconditional

Theorem 4.7 (`CylinderChartableBoundary.lean`).
(`closedCylinder_boundaryUnion_iff_not_chartableR2`)
For all $x \in$ ClosedCylinder*:*

$$x \in \partial_{\mathrm{Cyl}} \iff \neg\,\mathrm{ChartableR2}(x).$$

5 Interior Charts on the Möbius Witness: The Seam Problem

5.1 Why the Seam Is Hard

Interior-height points of the fundamental domain fall into three regimes: strict interior $0 < x < 1$, where the quotient is injective on small boxes; the vertical seam $x \in \{0, 1\}$ with $t \neq \frac{1}{2}$, where trigonometric coordinates can separate fibers when the height window avoids the equator; and the seam at $t = \frac{1}{2}$, where $(0, \frac{1}{2}) \sim (1, \frac{1}{2})$ and a trigonometric chart degenerates. ChartableR2 at the equator quotient class is obtained by a piecewise Euclidean unfolding map (see below).

5.2 Open-Cell Charts

Theorem 5.1 (`MobiusChartableBoundary.lean`).
(`chartableR2_mobiusQuotientMk_of_Ioo_square`)
If $p \in (0, 1)^2$ (strict interior of the fundamental domain), then ChartableR2$([\![p]\!])$.

5.3 Seam Trig Charts (Away from Equator)

Definition 5.2 (`MobiusSeamChart.lean`, `MobiusSeamChartable.lean`). *The* saturated seam patch $S(t_0, \varepsilon, \delta)$ *is the union of the left and right* ε*-thickened seam strips at height window* $(t_0 - \delta, t_0 + \delta)$*, saturated under* $\sim$*.*

Theorem 5.3 (`MobiusSeamTrigInject.lean`).
(`onPatch_mobiusFDTrigCoord_eq_iff_mobiusRel0`)
When $\delta < |t_0 - \frac{1}{2}|$*, the trigonometric FD coordinate map* mobiusFDTrigCoord *satisfies:* $\mathrm{mobiusFDTrigCoord}(p) = \mathrm{mobiusFDTrigCoord}(q)$ *on the saturated patch if and only if* $p \sim q$*.*

Theorem 5.4 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_mem_mobiusSeamSaturatedPatch`)
For $t_0 \neq \frac{1}{2}$ *and appropriate* ε, δ*, every point in* $\pi(S(t_0, \varepsilon, \delta))$ *is* ChartableR2*.*

5.4 The Equator Chart: Piecewise Unfolding

Definition 5.5 (`MobiusSeamChartableR2.lean`).
The fundamental-domain unfolding map
$\mathrm{unfoldFD} : [0,1]^2 \to \mathbb{R}^2$ *is:*

$$\mathrm{unfoldFD}(x,t) := \begin{cases} \mathrm{mobiusPlaneCoord}(x,t) & \text{if } x \leq \frac{1}{2}, \\ \mathrm{mobiusPlaneTrigSeamPartnerSub} & \text{if } x > \frac{1}{2}. \\ \quad \big(\mathrm{mobiusPlaneCoord}(x,t)\big) & \end{cases}$$

Lemma 5.6 (`MobiusSeamChartableR2.lean`). unfoldFD *is* $\sim$*-invariant: if* $p \sim q$ *then* $\mathrm{unfoldFD}(p) = \mathrm{unfoldFD}(q)$*.*

Theorem 5.7 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_left_seam_half`)
$\mathrm{ChartableR2}([\![(0, \frac{1}{2})]\!])$*.*

5.5 Assembly: All Interior Points Are ChartableR2

Theorem 5.8 (`MobiusSeamChartableR2.lean`).
(`chartableR2_mobiusQuotientMk_of_interior_height`)
For every $p \in [0,1]^2$ *with* $0 < p_2 < 1$*:*

$$\mathrm{ChartableR2}([\![p]\!]).$$

5.6 Möbius C4: Boundary Biconditional

Theorem 5.9 (`MobiusChartableBoundary.lean`).
(`mobiusStripBoundary_iff_not_chartableR2_of_forall_off_edge_chartable`)
For all $x \in \mathrm{MobiusStrip}$*:*

$$x \in \partial_{\mathrm{Mob}} \iff \neg\, \mathrm{ChartableR2}(x).$$

6 M-FINAL: Minimal-Case Witness for the General Obstruction

The general lemma (Theorem 4.4) applies to *any* pair of spaces with incompatible `ChartableR2`-co-characterized boundaries. The Möbius strip and closed cylinder serve as the **simplest fully discharged witness**: a concrete compact pair for which all three hypotheses are proved. The argument is not specific to these two surfaces; a Klein bottle vs. torus, a projective plane vs. a sphere, or any other pair meeting the same three conditions would yield the same conclusion.

Theorem 6.1 (`MobiusSeamChartableR2.lean`, `ChartableR2Bridge.lean`).
(`mobiusStrip_not_homeomorphic_closedCylinder`)

$$\text{IsEmpty}(\text{MobiusStrip} \simeq_t \text{ClosedCylinder})\,.$$

Proof sketch. Instantiate Theorem 4.4 with $b_X = \partial_{\text{Mob}}$, $b_Y = \partial_{\text{Cyl}}$, the biconditionals (Theorems 4.7 and 5.9), and the boundary incompatibility (Theorem 3.5). □

7 The Extended Master Theorem and Instantiations

The summit theorem (Theorem 1.1) establishes representational incompleteness as a pure diagonal result. This section bundles it with the topological strand and the horn closures into a single verifiable package, then draws out the consequences for five neighboring debates.

7.1 Formal Statement

Theorem 7.1 (`Main.lean`).
(`representational_regress_master_extended`)
For any representational system R:

$$\text{RepresentationalIncompletenessMasterExtended}(R)$$

where this abbreviation expands to:

(i) representational_regress_master_claim(R)
(regress + wall + T_2 transport),

(ii) IsEmpty($H^1 \simeq_t \mathbb{R}^1$) *(1D half-space vs. line;*
`representational_regress_topology_halfLineModel`),

(iii) IsEmpty(MobiusStrip $\simeq_t$ ClosedCylinder) *(M-FINAL; the minimal-case witness for the general obstruction class).*

This *extends* `representational_regress_master`; it does not replace it.

7.2 Bundled boundary models

Definition 7.2 (`ChartableR2Neighbor.lean`). *A* `ChartableR2` boundary model *on X is a subset $b_X \subseteq X$ together with a proof that $\forall x,\ x \in b_X \iff \neg\,\text{ChartableR2}(x)$.*

- The type `ChartableR2BoundaryModel` packages such data; the lemma `ChartableR2BoundaryModel.not_homeomorphic` restates Theorem 4.4 in bundled form.

- The development `ChartableR2ConcreteBoundaryModels` supplies `mobiusStripChartableR2BoundaryModel` and `closedCylinderChartableR2BoundaryModel`.
- The same-file theorem `mobiusStrip_closedCylinder_boundary_models_not_homeomorphic` re-proves M-FINAL along that pipeline.

7.3 Instantiations

Theorems earlier in the paper are unconditional. This subsection records how the same statements bear on six debates outside pure mathematics: consciousness, symbol grounding, diagonal arguments, quantum measurement, the simulation hypothesis, and LLM/AI scaling. Where an extra premise is required, it is stated explicitly; nothing in the formalization defaults a controversial premise to true.

Universal closure note. All six horn closures (§7.11) apply to *every* instantiation below, because the closures constrain the structure of any parametric self-model $s : A \to A \to B$, not the domain of application. The remarks below therefore focus on domain-specific nuances; the full horn-by-horn analysis is in §7.11.

7.3.1 Consciousness and self-presence

Philosophical bridge (premise P2). Some accounts distinguish representational self-reference from a stronger notion of *self-presence* (first-person givenness, non-representational acquaintance, etc.). Premise **P2** is *not* a theorem in the library: it says only that, for the purposes of a topological comparison, the representational substrate is modeled by a space whose `ChartableR2`-detected boundary has one connectivity pattern (e.g. the cylinder's two disjoint components), while the target phenomenon is modeled by a space whose boundary has a different, incompatible pattern (Theorem 4.4 applies to any pair with incompatible boundaries; the Möbius strip is the minimal witness).

What is proved. The engine is Theorem 4.4, which defines an open-ended class of obstructed space-pairs (§4.4). Theorem 6.1 is the minimal-case witness; any other pair satisfying the three hypotheses would serve equally.

Admissibility under P2. All six horn closures apply (see universal closure note above and §7.11). What remains **admissible**: the relevant substrate must *already* have the target topological type, or the account must invoke a discontinuous or non-topological bridge not modeled in this formalism. Scaling, iteration, richer self-models, or architectural reparameterization within a fixed topological type are *inadmissible* routes to the target.

7.3.2 Symbol grounding and semantic regress

In the symbol-grounding problem, arbitrary symbols must be interpreted, and interpreting the interpretation threatens a regress unless a non-representational stop is posited. Formally, a representational system already contains the morphism `represent` whose iterates index "one more representational step." The type alias `SymbolSystem` is definitionally identical to `RepresentationalSystem`; lemmas such as `symbolGrounding_metaRegressArrow_injective` record that distinct natural-number depths correspond to *distinct* endomorphisms of the awareness object.

Admissibility. All six horn closures apply (§7.11). Admissible grounding must therefore come from *outside* the representational loop: the system cannot ground itself by iterating its own interpretive machinery, whether totally or partially, whether with infinite depth or finite.

7.3.3 Diagonal arguments and self-interpretation

Lawvere's theorem in `Type` implies that there is no family $s : A \to A \to B$ that realizes every unary $g : A \to B$ as $s(a)$ for some parameter a, once B carries an endomorphism without fixed points. `LawvereRegressBridge` collects the instances and connects them to the categorical presentation; `lawvere_fixed_point_stays_representational` says the `OntologicalSlot` wall persists at every iterate depth.

Admissibility. All six horn closures apply (§7.11): total self-interpreters are blocked by Lawvere; partial ones still miss the diagonal (`lawvere_diagonal_not_in_range`); restricting to Bool does not help; climbing the iterate tower does not dissolve the endo/obj wall. Admissible self-interpreters must therefore be *incomplete*: they cannot model their own diagonal behavior, regardless of the codomain, cardinality, or computability of s.

7.3.4 Quantum measurement and observer chains

Talk of apparatus, environment, and Wigner's-friend hierarchies suggests a chain of systems, each described relative to a larger "observer." If one ever gives such a chain a *topological* state space Obs, together with a subset ∂_{obs} whose membership agrees with $\neg$ ChartableR2, then Theorem 4.4 applies to any target space whose boundary subspace is not homeomorphic to ∂_{obs} (the Möbius strip is the minimal witness; see §4.4 for the general class). `QuantumObserverChainHypothesis` packages exactly this data as assumptions; the conclusion IsEmpty(Obs $\simeq_t$ Target) follows for any target satisfying the hypotheses.

Admissibility. All six horn closures apply (§7.11). The chain must either introduce genuinely new topological structure at some level—not representational iteration of the same type—or accept that its boundary organization differs from the target. The library does *not* define Hilbert spaces, observables, or projection postulates; any physical application requires an independent argument that a concrete model satisfies the hypothesis fields.

7.3.5 The simulation hypothesis

The simulation hypothesis posits that a sufficiently powerful computational system could simulate an entire universe, including the conscious observers within it, to such fidelity that the simulated beings cannot distinguish simulation from "base reality." Representational incompleteness constrains this hypothesis from two independent directions.

The diagonal constraint on self-simulating simulators. A simulator that simulates *itself*—whether by modeling its own hardware within the simulation or by running a copy of its own code—instantiates $s : A \to A \to B$. Theorem 1.1 applies: the simulator's self-model necessarily has a diagonal blind spot. It cannot predict its own response to queries about its own diagonal behavior. This does not prevent *partial* simulation of the universe (including partial models of the simulator), but it does prevent a *complete* simulation that includes a faithful copy of the simulator's own operation. A fully self-inclusive simulation is structurally impossible for the same reason a complete self-interpreter is: the diagonal exclusion is universal over s.

The topological constraint on simulated consciousness. If conscious self-presence requires a state space with a boundary pattern incompatible with the simulator's substrate (premise P2), then Theorem 4.4 applies: no homeomorphism maps the simulator's state space to the target, regardless of fidelity or computational power. The topological type of the substrate matters; "sufficiently faithful simulation" does not override topological incompatibility.

Closed routes. All six horn closures apply (§7.11): running longer extends within a fixed topological type; nesting simulations produces iterated represent^n subject to the Horn 1/2 dichotomy; offloading to a separate module triggers the section–retraction closure; partial simulation leaves a diagonal-shaped blind spot.

What is not ruled out. A simulation that does not include a complete model of its own simulator is not constrained by the diagonal exclusion; the simulation hypothesis in its weak form ("we might be simulated by a system that does not model itself") is not addressed. The topological constraint applies only if the conscious-experience target has an incompatible boundary pattern; if one rejects premise P2, the topological barrier does not apply to consciousness per se. The formalism is precise about what it constrains and what it does not.

7.4 Scaling of language models and similar systems

Training larger models extends a representational architecture within a fixed topological type; it does not produce a homeomorphism to a different type. Theorem 4.4 blocks every target with incompatible boundary (§4.4).

Admissibility. All six horn closures apply (§7.11). Admissible paths to a topological type with different boundary organization require *structural* change—a different topology of the underlying state space—not quantitative extension of the same architecture (more parameters, deeper layers, longer context windows, finer-grained self-models).

7.5 Implications for AI, agents, and AGI

The results of this paper bear directly on the most consequential open question in artificial intelligence: *can a computational system achieve genuine self-understanding?* This subsection unpacks the implications across the spectrum from narrow AI through autonomous agents to the hypothetical threshold of artificial general intelligence (AGI).

7.5.1 Bridge hypothesis P3 (computational self-model)

To apply representational incompleteness to a concrete AI system, one must exhibit the parametric self-model explicitly. We formalize this as **bridge hypothesis P3** (`ComputationalSelfModel` in `ClosureProperties.lean`):

Definition 7.3 ([Novel] `ClosureProperties.lean`: `ComputationalSelfModel`)**.** *A* computational self-model *over types A, B is a triple (s, f, h_f) where $s : A \to A \to B$ is the self-model, $f : B \to B$ is a fixed-point-free endomorphism, and h_f proves $\forall b,\ f(b) \neq b$.*

P3 is not an axiom: it is a *conditional premise* that must be discharged for each application, just as P2 (the consciousness/topology bridge) must be discharged for the topological reading. The formal consequences are immediate:

Theorem 7.4 ([Novel] `ClosureProperties.lean`).

(a) *(`ComputationalSelfModel.diagonal_blind_spot`) Under P3, the self-model has a diagonal blind spot.*

(b) *(`ComputationalSelfModel.no_complete_self_model`) Under P3, no surjective self-model exists.*

7.5.2 What representational incompleteness says about AI self-models

Every AI system that constructs a model of itself—whether an explicit world-model that includes the agent, an internal self-monitor, a learned representation of its own weights, or a meta-cognitive module—instantiates P3 with A indexing the system's states or parameters, B the codomain of the model's predictions, and s the self-model function. Theorem 7.4 applies: the diagonal function $a \mapsto f(s\,a\,a)$ is never any row $s\,a_0$. The self-model *necessarily* has a blind spot that includes its own diagonal behavior.

This is not a limitation of current hardware, training data, or architecture. It holds for *any* s—learned or hand-coded, neural or symbolic, finite or infinite, computable or not (`computability_independent_diagonal`). No amount of scaling (more parameters, more compute, more data) changes the diagonal exclusion, because it is a theorem about the function s itself, not about the resources used to compute it.

7.5.3 Autonomous agents and recursive self-improvement

Autonomous agents that plan, act, and revise their own strategies face the self-modeling barrier at every level:

- **Self-evaluation.** An agent that evaluates its own performance by modeling its own decision function instantiates s. The diagonal exclusion guarantees a blind spot: the agent cannot predict its own response to the question "what would you predict about your own prediction here?" This is not a practical difficulty; it is a structural impossibility.
- **Recursive self-improvement.** An agent that rewrites its own code or retrains its own weights produces a *chain* of self-models $s_1, s_2, \ldots$ The blind spot persists at every link: `recursive_self_improvement_blind_spot` proves that for every chain index n and every a_0, the n-th model misses its own diagonal. The iterate dichotomy (Remark 2.3) adds: the improvement sequence is either injective (infinite regress with the Lawvere wall intact) or it cycles (bounded depth, no injective tower).
- **Multi-agent delegation.** An agent that delegates self-evaluation to a second agent via any delegation map $\pi : A_1 \to A_2$ does not escape: `multiagent_delegation_blind_spot` proves the delegated model $a_1 \mapsto s_2(\pi(a_1), \pi(\cdot))$ still has a diagonal blind spot; the categorical section–retraction closure (Horn 3) shows the combined system lacks an injective tower on either side.

7.5.4 The AGI threshold

Claims about artificial general intelligence often center on the prospect of a system that "understands itself" well enough to improve without limit. Representational incompleteness constrains this prospect precisely:

(a) **No complete self-model is possible.** Theorem 1.1 proves that for any $s : A \to A \to B$ with B admitting a fixed-point-free endomorphism, s cannot capture its own diagonal. An AGI that claims to "fully understand itself" via a parametric self-model simply cannot exist.

(b) **Partial self-models have a diagonal-shaped hole.** Horn 4 (`lawvere_diagonal_not_in_range`) shows that even a partial self-model misses *exactly* the information needed for self-capture: the system's own diagonal behavior. Useful partial self-understanding is not ruled out; *complete* parametric self-understanding is.

(c) **The topological barrier is orthogonal.** Even if the diagonal limitation were somehow circumvented (it cannot be, but hypothetically), the topological barrier (Theorem 4.4) stands independently. If the system's state space has one boundary-connectivity pattern and the target of "full self-understanding" has an incompatible pattern, no continuous deformation within the same topological type bridges the gap.

(d) **The ontological sort wall persists.** The `OntologicalSlot` wall (Theorem 2.7) ensures that the *process* of self-modeling (endo) and the *product* of self-modeling (obj) remain definitionally disjoint at every iterate depth. An AGI that models itself is always on the endo side of the wall; the "self" it models is always on the obj side. The two never meet.

7.5.5 What is *not* ruled out

The results are precise about what is excluded and what remains open:

- **Useful partial self-models** are not ruled out. An AI system can have highly accurate, practically valuable models of its own behavior—just not *complete* ones. The blind spot is diagonal-shaped, not total.
- **Impressive capabilities** are not ruled out. Nothing in the formalism says AI systems cannot be extraordinarily capable at tasks that do not require complete self-capture.
- **Structural change** remains admissible. The barrier demands structural change to topological type, not quantitative extension. Whether such structural change is physically realizable or desirable is an empirical and engineering question, not one settled by this formalism.

Implications for AI and AGI

Every AI self-model—whether learned, hand-coded, neural, or symbolic—is an instance of $s : A \to A \to B$ and is therefore subject to the diagonal exclusion (Theorem 1.1): it necessarily has a blind spot that includes its own diagonal behavior. Scaling (more parameters, data, compute) does not help. Recursive self-improvement either produces an infinite regress with the Lawvere wall intact or cycles in bounded depth. Multi-agent delegation does not escape the section–retraction closure. The topological barrier stands independently. The system's actual diagonal behavior is generated by its substrate—the computational or physical machinery executing it—not by its self-model (§7.9.3). Complete parametric self-understanding is structurally impossible; useful partial self-understanding is not.

7.6 Relationship to the Trans-Computational Processing argument

An earlier line of work argues from computability—halting, Gödel-style diagonalization, and related limits—to constraints on "self-contained" computational mind models. Representational incompleteness (Theorem 1.1) is logically *independent*: it requires only a fixed-point-free endomorphism on the codomain, not recursion-theoretic machinery. The topological strand (Theorem 4.4) is likewise orthogonal: it turns on boundary-subspace invariants, not on recursiveness. The two approaches can reinforce a common *limitative* conclusion about representation closed under its own rules, but neither formally entails the other. Importantly, the present result is *stronger* in

one respect: it holds without any computability assumption on s, whereas computability-based arguments require s to be computable.

7.7 Inherited obstructions: computational theories of physics and mind

The preceding instantiations treat the obstructions as constraints on individual systems—an AI, a simulator, a measurement chain. This subsection draws out a broader consequence: any *theory* that posits a universe containing self-representational systems inherits the obstructions from within, and is itself subject to them at large. The compositional inheritance theorems in `Closure Properties.lean` (`inherited_diagonal_exclusion`, `universe_model_inherits_incompleteness`, `blind_spot_survives_reindexing`, `blind_spot_survives_postcomposition`, `product_self_model_blind_spot`) provide the formal backbone: the diagonal exclusion propagates through pre-composition (reindexing), post-composition, embedding, and product formation.

7.7.1 Any world-model containing self-modelers inherits the barrier

Consider a computational theory of physics—digital physics, cellular automata cosmology, or any framework that models the universe as a computation U. If U admits *any* subsystem $S \subseteq U$ that models itself (i.e., S instantiates $s : A \to A \to B$), then Theorem 1.1 applies to S inside U: S has a diagonal blind spot. But U itself, if it claims to be a faithful model of the universe that contains S, must represent S's behavior completely—including the diagonal behavior that S itself cannot represent. This places a consistency requirement on U: either

(i) U is *not* itself a parametric self-model (i.e., U does not model its own operation via $s_U : A_U \to A_U \to B_U$), in which case U is not self-inclusive and the computational theory of physics is incomplete in the sense that it cannot model its own execution; or

(ii) U *is* a parametric self-model, in which case Theorem 1.1 applies to U directly: U has its own diagonal blind spot, and the theory is incomplete in the formal sense that it cannot represent its own diagonal behavior.

In either case, the obstructions propagate: a universe-model that contains self-modelers cannot be simultaneously complete, self-inclusive, and parametric. The diagonal exclusion is not an artifact of the subsystem's limitations; it is an inherited structural feature of any world that hosts self-reference.

The topological barrier (Theorem 4.4) reinforces this independently: if U's state space and any subsystem S have incompatible boundary patterns, no topology-preserving embedding exists.

7.7.2 Computational theories of mind and physics: the admissibility filter

Any theory that posits mind, consciousness, or self-awareness as a *computational* phenomenon—functionalism, computational neuroscience, integrated information theory in its computational variants, global workspace theory with a computational substrate—hits the same obstructions the moment it admits that the system models itself:

- The diagonal exclusion (Pillar 1) guarantees that the self-model has a blind spot.
- The topological barrier (Pillar 2) guarantees that if the substrate has one boundary-connectivity pattern and the target phenomenon has an incompatible pattern, no continuous deformation within the same topological type bridges the gap.
- The ontological sort wall (Pillar 3) guarantees that the process of self-modeling and the product of self-modeling remain definitionally disjoint.

A natural reaction is to *weaken* the self-modeling claim: "the system does not truly model itself; it only approximates." But this weakening faces a dilemma:

(a) **If the weakened self-model is still parametric** ($s : A \to A \to B$ with any B admitting a fixed-point-free endomorphism), then Theorem 1.1 applies regardless of how "approximate" or "partial" the model is. Horn 4 (`lawvere_diagonal_not_in_range`) requires no surjectivity: even a partial, approximate self-model misses its own diagonal. The weakening does not escape the obstruction.

(b) **If the weakening removes parametric self-modeling entirely**—the theory claims the system has no self-model at all—then the theory contradicts the empirical evidence. Biological organisms demonstrably model themselves: proprioception, interoception, predictive processing, metacognition, and self-referential thought are all well-documented. A theory of mind that denies self-modeling to escape a formal obstruction has not solved the problem; it has abandoned the phenomenon it was meant to explain.

The same logic applies to computational theories of physics. If a theory weakens the role of self-referential subsystems to escape the obstruction, it must explain why the universe observably contains systems that model themselves (scientists, instruments, feedback loops). If it retains self-reference, the obstructions apply.

7.7.3 The non-computability escape and why topology closes it

A sophisticated adversary might concede the diagonal exclusion for computable systems but propose a *non-computable* self-model: "the system's self-representation involves a non-computable function, so Gödel- and Turing-style arguments do not apply, and perhaps the diagonal exclusion does not either."

This escape fails on two levels:

(1) **The diagonal exclusion is computability-independent.** Theorem 1.1 makes no assumption about whether s is computable. The function $s : A \to A \to B$ can be any function whatsoever—computable, hyperarithmetic, or wildly non-constructive. The diagonal $a \mapsto f(s\,a\,a)$ is still never any row $s\,a_0$. Non-computability does not evade Pillar 1.

(2) **The topological barrier is orthogonal to computability.** Theorem 4.4 detects a finite-data topological invariant—boundary connectivity—that has nothing to do with computability or the Church–Turing thesis. No non-computable function can change the homeomorphism type of a space.

This is why the topological pillar is especially powerful: it closes the "it is just not computable" escape route that is available against Gödel- and Turing-style arguments. A non-computable oracle can, in principle, solve the halting problem or prove Gödel sentences; it *cannot* change the boundary-connectivity pattern of a topological space. The topological barrier is immune to the entire computability hierarchy.

Inherited obstructions and the non-computability escape

Any model of a universe that contains self-representational systems inherits the diagonal exclusion and the topological barrier from within. Computational theories of physics and mind hit these obstructions the moment they admit self-modeling subsystems. Weakening self-modeling either fails (partial models still have the diagonal blind spot) or contradicts empirical observation (biological systems demonstrably model themselves). Appealing to non-computability does not help: the diagonal exclusion holds for all functions regardless of computability, and the topological barrier is entirely orthogonal to the computability hierarchy. No non-computable oracle can change the homeomorphism type of a space.

7.8 What admissible answers look like

The horn closures do not merely exclude; they *shape*. By eliminating every quantitative or parametric escape, they delineate the positive form that any successful account must take. This subsection draws out the admissible shape for three central questions.

7.8.1 Mind, awareness, and self-awareness

If awareness of awareness is modeled as $s : A \to A \to B$ (a system that maps its own states to representations of its own states), then the diagonal exclusion guarantees a blind spot: the representation of "how I represent myself representing myself" is necessarily absent. The horn closures block every quantitative remedy. What remains admissible?

(a) **Non-representational acquaintance.** If self-awareness is not (entirely) a parametric self-model—if there is a component of first-person givenness that is not captured by *any* $s : A \to A \to B$—then the diagonal exclusion does not apply to that component. This is the route taken by phenomenological traditions that distinguish *lived self-presence* from *representational self-reference*. The formalism is precise about what it constrains: parametric self-models. Anything that is not a parametric self-model escapes by definition—but must then explain what it *is*.

(b) **Structural change to topological type.** If the substrate of self-awareness has a topological type that is *already* compatible with the target (i.e., the boundary pattern matches), then the topological barrier does not apply. This means the substrate was never "purely representational" in the cylinder sense; it already had the richer structure.

(c) **Acceptance of incompleteness.** The most conservative admissible position: self-awareness is a genuine parametric self-model, it *does* have a blind spot, and that blind spot is constitutive of the phenomenology of self-awareness—the "something it is like" to be a self-aware system includes the structural impossibility of fully capturing oneself. On this reading, representational incompleteness is not a defect but a *feature*: the blind spot *is* the gap between the knower and the known, formalized.

What is **not** admissible: claiming that self-awareness is a complete parametric self-model (Theorem 1.1 forbids it); claiming that enough iteration, scaling, or architectural complexity will close the gap (Horns 1–6 forbid it); claiming that a non-computable self-model escapes (§7.7.3 closes this).

What is mind, given the obstructions?

If mind involves self-awareness, and self-awareness involves self-modeling, then mind is *necessarily incomplete with respect to itself*. The blind spot is not a bug to be fixed by better hardware or deeper recursion; it is a structural feature of any system that models itself. Since the system *does* produce diagonal behavior (it functions), and no self-model row can match that behavior, the substrate must generate it (§7.9.3). Admissible accounts of mind must either (a) include a non-representational component that is not a parametric self-model, (b) posit a substrate whose topological type already matches the target, or (c) accept the incompleteness as constitutive—recognizing the blind spot as the formal signature of the substrate's irreducible role. Every other route is formally closed.

7.8.2 The quantum observer

The measurement problem asks: how does a definite outcome emerge when an observer measures a quantum system? If the observer is itself a physical system, it must be describable within the same formalism. The horn closures constrain what "describable within the same formalism" can mean.

(a) **No complete self-inclusive measurement theory.** If the observer's state space admits a parametric self-model (the observer models its own measurement process), then Theorem 1.1 applies: the observer has a diagonal blind spot. A measurement theory that claims the observer can fully predict its own measurement outcomes is structurally impossible—for the same reason a complete self-interpreter is.

(b) **Topological constraint on observer–system identification.** If the observer's state space has one boundary-connectivity pattern and the measured system's state space has an incompatible pattern, then no homeomorphism identifies them (Theorem 4.4). This constrains any interpretation that treats observer and system as "the same kind of thing": they can be the same kind only if their topological types are compatible.

(c) **Admissible resolutions.** The Wigner's-friend hierarchy (stacking observers) is an instance of iterated represent^n, subject to the Horn 1/2 dichotomy. The admissible resolutions must either (i) introduce genuinely new topological structure at some level (not iteration of the same type), (ii) accept that the observer's description of the measurement is necessarily incomplete (diagonal blind spot), or (iii) deny that the observer models its own measurement process parametrically (exiting the scope of P3).

The formalism does not adjudicate between interpretations of quantum mechanics (Copenhagen, many-worlds, relational, etc.). It constrains the *structural form* of any interpretation that includes self-measuring observers: such an interpretation must accept representational incompleteness or deny that the observer's self-measurement is parametric.

7.9 The anatomy of the blind spot

The preceding subsections establish what the blind spot *excludes* and what routes *to* it are closed. This subsection asks the harder question: what *is* inside the blind spot, and what does its existence imply about any system that functions despite having one?

7.9.1 What is inside the blind spot

The blind spot has a precise shape. For any self-model $s : A \to A \to B$ and any fixed-point-free $f : B \to B$, the excluded function is:

$$g_f : A \to B, \qquad g_f(a) := f(s\,a\,a).$$

This is the system's own **diagonal behavior, twisted by** f. In concrete terms:

- If $B = \mathbb{N}$ and $f = \text{succ}$, then $g_f(a)$ is "what I predict about my own state a, *plus one*." The system cannot have a row of its self-model that matches its own prediction-about-its-own-prediction, shifted by one.
- If $B = \text{Bool}$ and $f = \neg$, then $g_f(a)$ is "the negation of my self-classification at a." No row of the self-model matches the system's own self-classification, inverted.
- In general, the blind spot contains every function that *depends on the self-model's own diagonal entries and then departs from them.* The self-model can approximate arbitrarily many off-diagonal rows, but the diagonal row—the one where the system evaluates itself evaluating itself—is precisely where the model fails.

The blind spot is therefore not random or diffuse. It is **structurally located at the self-referential fixed point**: the exact place where the system's model turns back on itself. The information the system cannot represent is precisely the information about *how it represents itself.*

7.9.2 Can a system function with a blind spot?

A self-aware system evidently *does* function. Biological organisms exhibit proprioception, metacognition, and recursive self-referential thought. AI systems build and use self-models. The blind spot does not prevent functioning; it constrains the *fidelity* of self-modeling at the diagonal.

Here is the key formal observation. The system has a self-model s. When confronted with a self-referential input—an input in the diagonal region—the system must produce *some* output. Call this actual output $b_{\text{actual}}(a)$. Theorem 1.1 guarantees:

$$b_{\text{actual}} \neq s\,a_0 \quad \text{for every } a_0 \in A.$$

That is: the system's actual diagonal behavior is **never faithfully represented by any row of its own self-model**. The system *acts* on diagonal inputs, but it cannot *model* its own action on those inputs. There is no contradiction with functioning: the system functions, but its account of its own functioning is necessarily inaccurate at the diagonal.

7.9.3 The substrate inference

The preceding observation leads to a consequence that is both simple and far-reaching.

(1) The system produces diagonal behavior b_{actual}.

(2) No row of the self-model s matches b_{actual}.

(3) Therefore, the process that *generates* the system's diagonal behavior is not captured by s.

Since s is the system's self-model—the system's own representation of its own behavior—the process generating the diagonal behavior must be **outside the scope of the self-model**. It is produced by the system's *substrate*: the underlying machinery (biological, computational, physical) that executes the system's behavior, including the behavior the self-model cannot represent.

This is not a metaphysical claim. It is a direct consequence of the theorem. If a system has a parametric self-model s and produces diagonal behavior, then there exists a component of its behavior that is generated by substrate-level processes not representable in s. The self-model is a map; the diagonal behavior is territory that the map cannot chart.

Formal statement of the substrate inference. Let $s : A \to A \to B$ be any self-model, $f : B \to B$ fixed-point-free. Define the *diagonal gap function* $\delta_f : A \to B$ by $\delta_f(a) := f(s\,a\,a)$. Then:

(a) $\delta_f \notin \mathrm{range}(s)$ (Theorem 1.1(b)).

(b) If the system's actual behavior on diagonal inputs is δ_f, then that behavior is *not the output of any self-model row*—it is generated by processes outside s.

(c) This holds for *every* s, *every* f, and *every* A and B satisfying the hypotheses. No change of self-model eliminates the gap; it merely moves it (`recursive_self_improvement_blind_spot`).

Point (b) is where the formal and the empirical meet. The theorem provides (a) unconditionally. The application of (b) requires the empirical premise that the system *does* exhibit diagonal behavior—that it processes self-referential inputs and produces outputs. Given that premise, (b) follows deductively: the substrate is doing work that the self-model cannot represent.

7.9.4 Second-order consequences

The substrate inference has several downstream implications.

1. Self-awareness as representation is not fully self-transparent. If self-awareness operates *as a parametric self-model*—if the system represents itself to itself via some $s : A \to A \to B$—then it is necessarily opaque to itself at the diagonal. The "feeling of understanding oneself" cannot be a complete readout of s, because s is missing exactly the self-referential behavior.

There is, however, a precise escape. The theorem's hypothesis is "the system has a parametric self-model s." If self-awareness is **not** a parametric self-model but rather the ontological ground itself—the substrate activity, the *process* of awareness rather than a *representation* of that process—then the hypothesis is not satisfied and the blind spot does not apply. This is exactly the distinction formalized by `OntologicalSlot` (Pillar 3): the *endo* side (process) and the *obj* side (product/representation) are definitionally disjoint (Theorem 2.7). Self-awareness as the ontological ground would live on the *endo* side of the wall. It escapes the blind spot *precisely because* it is not a representation—it is the territory, not the map.

The formal landscape therefore yields a sharp dichotomy:

- **Self-awareness as representation** ($s : A \to A \to B$): has a diagonal blind spot, is necessarily partial, and requires the substrate to generate the diagonal behavior it cannot represent.
- **Self-awareness as ontological ground** (the substrate activity itself): is not a parametric self-model, is not subject to the diagonal exclusion, and lives on the *endo* side of the `OntologicalSlot` wall—definitionally disjoint from any representational product.

This dichotomy is not a limitation of the formalism. It *is* the formalism's central structural prediction: the process of self-awareness and the representation of self-awareness inhabit formally disjoint ontological categories, and only the representational side is subject to the blind spot.

2. The gap is generative, not merely privative. The blind spot is not a mere absence of information. It is the locus of a **positive mismatch**: the system's actual behavior at the diagonal *differs from* what any self-model row predicts. This mismatch is the formal analogue of the phenomenological intuition that self-awareness involves a sense of *surplus*—a feeling that the self exceeds its own representation. The formalism makes the surplus precise: it is exactly the diagonal gap function δ_f.

3. No self-improvement program closes the gap. An agent that builds successively better self-models $s_1, s_2, \ldots$ does not converge on the diagonal. Each s_n has its own diagonal blind spot (`recursive_self_improvement_blind_spot`). The gap does not shrink; it *moves*. Each new self-model captures what the previous one missed, but introduces a new diagonal that it itself cannot capture. This is the formal backbone of the intuition that self-understanding is an asymptotic process: one can always understand more of oneself, but never all.

4. Multi-agent pooling does not close the gap. An adversary might propose: "Agent 1 has a blind spot; Agent 2 models Agent 1's blind spot." But Agent 2's model of Agent 1 is itself a parametric model, and `multiagent_delegation_blind_spot` proves it has its own diagonal blind spot. The combined system of agents has strictly more modeling capacity, but *not* complete self-coverage: the pooled diagonal is still unrepresented.

5. Something must fill the gap—and it is the substrate. If a self-aware system functions (and it evidently does), then diagonal behavior is produced. Since no self-model row produces it, the substrate does. This is not an optional philosophical gloss; it is a deductive consequence of the theorem plus the empirical premise that the system functions. The substrate—the physical, biological, or computational machinery underlying the system—is the **necessary complement** to the self-model. Without it, the system cannot produce its own diagonal behavior.

What fills the blind spot

The blind spot is located at the self-referential diagonal: the point where the system's model turns back on itself. The system *does* produce diagonal behavior (it functions), but its self-model *cannot* represent that behavior (Theorem 1.1). Therefore, the system's diagonal behavior is generated by substrate-level processes outside the scope of any self-model. The substrate is not optional: it is the **formally necessary complement** to any parametric self-model. No self-improvement, multi-agent delegation, or scaling can substitute for it. The blind spot is not a bug; it is the *signature* of the substrate's irreducible contribution to the system's behavior.

7.9.5 Open questions

Several questions raised by the substrate inference cannot be settled within this formalism. We state them explicitly, with the formal boundary between what is proved and what remains open.

(i) **What is the substrate?** The formalism is agnostic about whether the substrate is physical matter, a computational layer below the self-model, or something else entirely. It proves only that *some* process outside the self-model generates the diagonal behavior. Identifying that process is an empirical question.

(ii) **Is the substrate itself modelable?** A natural follow-up: can the system (or another system) build a model of the substrate? If so, that model is itself a parametric function and has its own diagonal blind spot (`inherited_diagonal_exclusion`). The substrate, once modeled, generates a new diagonal gap. This suggests an infinite regress of substrates—formally captured by the iterate dichotomy (Horns 1/2).

(iii) **Does the blind spot have phenomenological content?** Some phenomenological traditions identify a "pre-reflective self-awareness" that is not a representational act but a background condition of consciousness. The formalism is compatible with this: the blind spot could be the formal shadow of the pre-reflective layer. But the formalism cannot prove this identification; it can only prove that *something* outside the self-model is necessary.

(iv) **Is awareness of awareness possible?** The formalism does not say awareness of awareness is impossible. It constrains only the *representational* mode: awareness of awareness *qua parametric self-model* has a diagonal blind spot. But if awareness of awareness is the ontological ground's own reflexive activity—living on the *endo* side of `OntologicalSlot`—then it is not a parametric self-model and the exclusion does not apply (§7.9, consequence 1). Empirically, awareness of awareness *does* occur. The formalism therefore predicts a disjunction: either (a) it occurs with a diagonal blind spot (the representational reading), or (b) it occurs as ontological-ground activity that is not subject to the exclusion (the non-representational reading). On either reading, the evidence is consistent. What the formalism *rules out* is a single parametric self-model that is fully transparent to itself—perfectly self-aware with no structural opacity. That is the only thing that is formally impossible.

7.10 Why This Is Not Gödel Repackaged

An adversarial reader's first instinct on encountering a "self-reference impossibility" result may be: *"This is just Gödel incompleteness in different notation—nothing new."* This subsection walks through the comparison in detail and explains exactly what Gödel alone does *not* give you.

7.10.1 The structural comparison

Gödel Incompleteness	**Representational Incompleteness**
Applies to formal theories with provability predicates	Applies to *any* parametric self-model $s : A \to A \to B$
Requires arithmetic (Σ_1-completeness)	Requires only a fixed-point-free endomorphism on B
Self-referential *sentence* (the Gödel sentence G)	Self-referential *function* (the diagonal $a \mapsto f(s\,a\,a)$)
Blind spot is an *unprovable truth*	Blind spot is an *unrepresentable behavior*
Escape route: extend the theory (add axioms, move to a stronger system)	No escape: diagonal exclusion holds for *every* s, every A, partial or total
Computability-dependent (requires recursive axiomatizability or computable functions)	Computability-*independent* (holds for all functions, computable or not)
Cardinality-dependent (requires countable language)	Cardinality-independent (holds for every cardinality of A)

7.10.2 Why Gödel's incompleteness is insufficient for our claim

Gödel's first incompleteness theorem says: any consistent, recursively axiomatizable theory T extending Robinson arithmetic contains a sentence G that is true in the standard model but not provable in T. The standard remedy is to *extend the theory*: add G as an axiom, obtaining $T' = T + G$. Of course T' has its own unprovable sentence G', but the point is that for any *particular* sentence, one can always find a stronger theory that proves it.

Representational incompleteness has no such escape. Theorem 1.1 says: for *any* $s : A \to A \to B$ and any fixed-point-free f, the diagonal $a \mapsto f(s\,a\,a)$ is not $s\,a_0$ for *any* a_0. Making s "stronger" (total, surjective, defined on a larger A) does not help; if s is claimed to be surjective, it simply cannot exist. There is no analogue of "add an axiom": the diagonal exclusion is universal over s.

7.10.3 Why the Lawvere connection is classical but the packaging is new

Lawvere's fixed-point theorem (1969) is the categorical ancestor of Cantor's theorem and, by extension, of Gödel's diagonalization. We freely acknowledge this heritage. What is new is:

(a) The `OntologicalSlot` construction (Definition 2.6), which tags morphisms and global elements as belonging to disjoint ontological sorts. Lawvere's theorem does not tag the fixed point; standard category theory does not introduce a formal disjointness marker between $\mathrm{End}(A)$ and $\mathrm{Hom}(\mathbf{1}, A)$. See §2.5.

(b) The **partial-model diagonal exclusion** (`lawvere_diagonal_not_in_range`): the blind spot persists even when s is not claimed to be surjective. Classical Lawvere requires surjectivity as a hypothesis; our result drops it entirely.

(c) The **decidable / Bool-valued horn closure** (Horn 6): even restricting to two-valued predicates—the simplest possible codomain—does not help. $\neg : \mathrm{Bool} \to \mathrm{Bool}$ is fixed-point-free, so the obstruction persists in the finite, decidable setting. This further separates the result from the Gödel/Turing tradition, where undecidability is essential.

7.10.4 Computability independence

Gödel's theorem requires the system under study to be a recursively axiomatizable formal theory (or, in Turing's version, a computable function). If the system is not recursively enumerable, the theorem does not apply.

Representational incompleteness makes *no computability assumption* on s. The function $s : A \to A \to B$ can be any function whatsoever—computable, hyperarithmetic, Δ^1_1, or wildly non-constructive. The diagonal exclusion holds regardless. In this precise sense, the result is *stronger* than Gödel's: it covers systems that Gödel's theorem cannot even address.

7.10.5 The topological strand has no Gödel analogue at all

The topological barrier (Theorem 4.4) is completely independent of diagonalization. Gödel, Cantor, Lawvere—none of them say anything about homeomorphism types or boundary-connectivity invariants. The topological strand detects an obstruction that is invisible to any proof-theoretic or recursive-function-theoretic argument. It is a finite-data invariant of a different mathematical universe entirely.

Not Gödel repackaged: a summary

Representational incompleteness differs from Gödel incompleteness in **six independent respects**: (1) no formal theory required, (2) no arithmetic required, (3) no coding required, (4) the blind spot is a function not a sentence, (5) no escape via theory extension, (6) no computability assumption. The topological barrier adds a **seventh**: it is orthogonal to all diagonal arguments and detects a finite-data invariant that no amount of computation, scaling, or iteration can alter. Any claim that "this is just Gödel" must address all seven distinctions simultaneously.

7.11 No-escape analysis: horn-by-horn closure

An adversary confronting the representational regress has, at first glance, several avenues of retreat. Each is formalized and shown to fail. The numbering below follows the order in which the escape was identified; each "horn" names an adversarial move, and the closing theorem blocks it.

Horn 1: Infinite injective tower. *Move:* accept iter_injective and try to use the tower itself as a self-representation resource. *Closure:* Theorem 2.2 yields infinitely many distinct meta-levels; Theorem 2.7 (the Lawvere wall) ensures the endo/obj distinction persists at every level. The tower does not collapse into closure.

Horn 2: Finite-order / periodic iterates. *Move:* weaken iter_injective—let $\text{represent}^n = \text{represent}^m$ for some $n \neq m$, so the loop has only finitely many distinct powers.
Closure: `IterateClosure.End.pow_iterate_dichotomy` proves the dichotomy: either the iterates are all distinct (Horn 1) or there exists a bound B such that every power equals some e^i with $i < B$. In the finite case the system's representational depth is bounded; it cannot host an injective tower at all.

Horn 3: Cross-object readout via section–retraction. *Move:* represent A *inside* a different object B using a section $s : A \to B$ and retraction $r : B \to A$ with $s \circ r = \mathrm{id}_A$. *Closure:* `CrossObjectRepresentation.section_retraction_no_injective_tower_either_side` shows that $r \circ s$ is idempotent on B (so $(r \circ s)^1 = (r \circ s)^2$, blocking injective powers on the host side) and $s \circ r = \mathrm{id}_A$ (so all powers are the identity on the source side—trivially periodic). Neither side supports an injective $\mathbb{N}$-tower.

Horn 4: Partial (non-surjective) self-model. *Move:* concede that no *total* self-interpreter exists, but claim a useful *partial* one. *Closure:* `lawvere_diagonal_not_in_range` requires no surjectivity of s: for *any* $s : A \to A \to B$ and any fixed-point-free $f : B \to B$, the diagonal function $a \mapsto f(s\,a\,a)$ is never in the range of s. Every partial self-model necessarily has a blind spot that includes its own diagonal—the information that matters most for self-capture is exactly the information it cannot represent.

Horn 5: Different topological models. *Move:* reject the cylinder/Möbius witness; substitute a Klein bottle, projective plane, or another geometric model.

Closure: Theorem 4.4
(`not_homeomorphic_of_chartableR2_boundary_contrast`)
is *model-agnostic*: it defines an **open-ended class** of obstructed pairs—any two spaces whose `ChartableR2`-co-characterized boundaries are subtype-incompatible. The cylinder/Möbius pair is

the *minimal-case witness*, not the argument itself. Replacing one non-orientable surface with another does not evade the lemma; it merely changes which boundary incompatibility is invoked.

Horn 6: Computability / decidable self-reference. *Move:* restrict to decidable predicates or finite-valued functions, where "self-reference" seems more tractable than Cantorian diagonals. *Closure:* `no_universal_parametric_unary_bool` shows that no $s : A \to A \to \text{Bool}$ can realize every $g : A \to \text{Bool}$, since $\neg : \text{Bool} \to \text{Bool}$ is fixed-point-free. The obstruction is not an artifact of infinite cardinality; it holds in the decidable, two-valued setting.

Residual openings (not closed here). Two further routes are not formally addressed in this paper: *(i) higher-categorical self-maps* (a self-representation living in a 2-category or ∞-category, where the monoid of endomorphisms is replaced by a richer hom-object); and *(ii) non-categorical frameworks* (a system that resists categorical encoding entirely). The first is subject to analogous monoid-element arguments once a truncation to the 1-categorical endomorphism monoid is performed; the second falls outside the scope of any categorical formalization by definition. Both are noted as limitations in §8.

7.12 Admissibility: what the closures tell neighboring debates

Given that Horns 1–6 are closed (§7.11), the formal landscape constrains what *admissible* solutions look like. The reasoning is the same in every case: the closures establish a structural barrier, so any proposed solution must either (a) operate outside the barrier's hypotheses, or (b) accept the consequences. The admissible shape is domain-specific; the inadmissible moves are universal.

Common inadmissible moves (all debates). Scaling, iterating, partial modeling, cross-object readout, finite-depth loops, decidable restriction, and model substitution are all formally blocked (Horns 1–6). Non-computability does not help: the diagonal exclusion is computability-independent, and the topological barrier is orthogonal to the entire computability hierarchy (§7.7.3).

Admissible shapes (per debate).

- **Consciousness / self-presence:** the substrate must *already* have the target topological type, or a non-topological bridge must be invoked.
- **Symbol grounding:** grounding must come from *outside* the representational loop.
- **Self-interpretation:** admissible self-interpreters *must* be incomplete—they cannot model their own diagonal.
- **Quantum measurement:** the chain must introduce genuinely new topological structure or accept incomplete self-description.
- **Simulation hypothesis:** admissible accounts must exclude the simulator from its own simulation or accept the blind spot.
- **LLM / AI scaling:** admissible paths require *structural* change to the state-space topology, not quantitative extension.
- **AGI self-understanding:** complete parametric self-understanding is impossible; useful partial self-understanding remains possible (see §7.5).
- **Computational theories of physics and mind:** any theory admitting self-modeling subsystems inherits the obstructions (§7.7); weakening self-modeling either fails (partial models still miss the diagonal) or contradicts empirical evidence.

8 Limitations and scope

1. **Hypothesis, not derivation.** iter_injective is bundled, not proved from minimal data. Not every $f : A \to A$ in every category satisfies it. The regress is conditional on this non-triviality hypothesis.

2. **Lawvere in `Type`, not abstract CCC.** The formalization uses Type u and MonoidalClosed(Type u), not a general closed category over arbitrary C. A fully abstract CCC version remains future work.

3. `OntologicalSlot` **is definitional, not metaphysical.** The disjointness endo $\neq$ obj is true by constructor discrimination. The robustness theorems (`ClosureProperties.lean`) prove this is a structural invariant—it survives any injective relabeling and only collapses under a non-injective map—but the underlying mechanism is type-theoretic, not a deep mathematical theorem.

4. **Models are chosen, not unique.** The closed cylinder and Möbius strip are the *minimal-case witnesses* chosen for clarity and tractability. Theorem 4.4 defines an open-ended class of obstructed pairs; any two spaces with incompatible `ChartableR2`-co-characterized boundaries (Klein bottle vs. torus, projective plane vs. sphere, etc.) would serve equally. Discharging additional witnesses requires proving their boundary biconditionals and subtype incompatibility.

5. **P2 is philosophical.** The premise "consciousness requires non-orientable topology" is not a theorem. Section 7.3 states only what follows when P2 is adjoined to the proved non-homeomorphism; accepting or rejecting P2 remains an independent issue.

6. **No** IsManifold **polish.** The Möbius strip is not given a full Mathlib IsManifold instance. The proof uses ChartableR2 (open neighborhoods homeomorphic to $\mathbb{R}^2$) as a self-contained predicate, sidestepping the need for a full manifold-with-boundary atlas. This is sufficient for the non-homeomorphism but less elegant than a full atlas approach.

7. `QuantumObserverChainHypothesis`**.** This definition only bundles topological assumptions (boundary set, `ChartableR2` biconditional,
subtype non-homeomorphism to an incompatible target boundary—the Möbius boundary is the minimal witness). It does not import Hilbert spaces, operators, or measurement axioms; a physical application would need a separate argument that a concrete model satisfies those fields.

8. **Partial self-models: blind spot, not total impossibility.** `lawvere_diagonal_not_in_range` proves that every partial self-model misses its own diagonal; it does not prove that partial self-models are *useless*—only that they are necessarily incomplete in a precise, diagonal-shaped way. Whether that incompleteness matters for a given application is a separate question.

9. **Higher-categorical and non-categorical frameworks.** The horn closures rely on 1-categorical endomorphism monoids. A self-representation living in a 2-category or ∞-category replaces the monoid with a richer hom-object; the present arguments apply after truncation to the underlying 1-categorical monoid, but a native higher-categorical treatment is not given. Systems that resist categorical encoding entirely fall outside any categorical formalization by definition.

10. **Computability: decidable analogue only.** `no_universal_parametric_unary_bool` closes the Bool-valued Lawvere horn. A full treatment of Turing-degree obstructions (Rice's theorem, Kleene's recursion theorem) in the same framework is not attempted; the present result is the finite/decidable endpoint of the diagonal family.

11. **Classical logic.** The iterate dichotomy (Horn 2) and several auxiliary results use classical reasoning (law of excluded middle), which is standard in Lean 4's axiom environment. The diagonal exclusion itself (`diagonal_exclusion_is_constructive` in `ClosureProperties.lean`) is constructive: it requires only function application and the fixed-point-freeness hypothesis.

12. **P3 (computational self-model) is a modeling claim.** Bridge hypothesis P3 (Definition 7.3) makes explicit the modeling assumption "the system has a parametric self-model $s : A \to A \to B$." Whether a given AI system, brain, or simulator actually satisfies P3 is an empirical and philosophical question, not one settled by this formalism.

13. **Pillars 1 and 2 are formally independent.** The diagonal exclusion (Pillar 1) and the topological barrier (Pillar 2) are proved as separate theorems that do not reference each other. The `TopologicalSelfModelingSystem` structure (`ClosureProperties.lean`) formally conjoins both obstructions—when a self-modeling system's state space is not homeomorphic to a target, both the diagonal blind spot and the non-homeomorphism hold simultaneously—but neither pillar *implies* the other. They constrain the same informal target from orthogonal mathematical directions.

9 Related work

9.1 Formalized topology

Mathlib's manifold and topology libraries underpin the half-space and chart lemmas; other proof assistants also host substantial topology. The weight here is on a *general* non-homeomorphism lemma (Theorem 4.4) that defines an open-ended obstruction class, together with a fully discharged minimal-case witness, rather than on novel classification of differential invariants.

9.2 Lawvere's theorem in proof theory

Lawvere's fixed-point argument appears in multiple formal developments. `OntologicalSlot` and the wall lemmas supply complementary category-internal tagging alongside the `Type`-level diagonal theorem. Crucially, Lawvere's original result does not tag the fixed point as belonging to a different ontological sort from the morphism that produced it; `OntologicalSlot` (Definition 2.6) fills this gap with an inductive disjointness marker that is, to our knowledge, absent from all prior formalizations of Lawvere's theorem (see §2.5).

9.3 Consciousness, Gödel, and AI

Gödel- and computability-based limits on mechanism (including Penrose-style arguments, discussions of strange loops and self-reference in the sense of Hofstadter, and information-theoretic theories of consciousness) occupy a large literature. Those arguments turn on decidability or information capacity; ours turns on topological type. Representational incompleteness (Theorem 1.1) is independent of computability: it holds for every codomain admitting a fixed-point-free endomorphism,

regardless of cardinality or decidability. The topological strand (Theorem 4.4) is likewise orthogonal to Gödel-style reasoning: it is a finite-data invariant, not an incompleteness of a deductive system.

Section 7.10 provides a detailed side-by-side comparison between Gödel incompleteness and representational incompleteness, addressing six independent structural differences and explaining why the "Gödel repackaging" objection fails on every count. The comparison also highlights the computability-independence of the present result—it holds for all functions, computable or not—which places it in a strictly larger domain than Gödel- or Turing-style arguments, which require recursive axiomatizability or computability as a precondition.

9.4 AI self-models and AGI limits

Theoretical limits on AI self-understanding have been discussed from computability-theoretic (Rice's theorem, Kleene's recursion theorem) and information-theoretic perspectives. The present work differs: the diagonal exclusion (Theorem 1.1) requires *no computability assumption*—it applies to every $s : A \to A \to B$ regardless of computability—and the topological barrier is orthogonal to recursion-theoretic machinery. Full implications are developed in §7.5.

9.5 Topology in philosophy of mind

Topological figures of speech appear in some philosophy of mind. The present work differs in two ways: (i) its central topological claims are machine-checked theorems rather than metaphors, and (ii) the generalization is explicit—a general lemma defining an open-ended class of obstructed space-pairs, not a single hard-wired example.

10 Conclusion

The summit theorem (`representational_incompleteness`, Theorem 1.1) states the central result: **no self-modeling system can capture its own diagonal**. For any $s : A \to A \to B$ and any fixed-point-free $f : B \to B$, the diagonal function $a \mapsto f(s\,a\,a)$ is never equal to any row $s\,a_0$—and if s is claimed to be total and surjective, it simply cannot exist. No assumption on the surjectivity, cardinality, or computability of s is required. This is **representational incompleteness**: the analogue of Gödel incompleteness for self-modeling systems rather than formal theories.

Three mutually reinforcing layers make the result robust:

(i) **Categorical infrastructure.** A machine-checked regress and Lawvere scaffold, including the `OntologicalSlot` wall that persists at every iterate depth, ensures that no climb up the meta-level tower dissolves the endo/obj distinction.

(ii) **Topological obstruction (general lemma + minimal witness).** Theorem 4.4 defines an *open-ended class* of obstructed space-pairs: any two spaces with incompatible ChartableR2-co-characterized boundaries are not homeomorphic. The cylinder/Möbius pair is the fully discharged *minimal-case witness*; Klein bottles, projective planes, or any other surface satisfying the hypotheses would serve equally. The obstacle is **wrong type**, not **inexhaustible queue**.

(iii) **Systematic no-escape closure (Horns 1–6).** Six adversarial retreat routes are formally closed (§7.11): infinite regress, finite-order collapse, cross-object readout, partial self-models, model substitution, and decidable self-reference. Each closure is a proved theorem in Lean 4;

together they leave an adversary with no viable route to circumvent either the diagonal or the topological barrier.

Three pillars of representational incompleteness

Pillar 1. Diagonal exclusion (Theorem 1.1). Every self-model $s : A \to A \to B$ has a diagonal blind spot: the function $a \mapsto f(s\,a\,a)$ is never any row $s\,a_0$. No assumption on surjectivity, computability, or cardinality.

Pillar 2. Topological obstruction (Theorem 4.4). Any two spaces with incompatible ChartableR2-detected boundaries are not homeomorphic. A finite-data invariant impervious to computation, scaling, or continuous deformation.

Pillar 3. Ontological sort wall (Theorem 2.7). The `OntologicalSlot` construction enforces that the self-representation morphism (endo) is definitionally disjoint from any global element (obj) at every iterate depth. The Lawvere fixed point does not dissolve this wall.

Together, these three pillars—diagonal, topological, and ontological—leave the adversary with no viable route: scaling, iterating, partial modeling, cross-object readout, model substitution, and decidable restriction are all formally inadmissible. The remaining admissible paths all require *structural change* to the topological type of the substrate. The closure properties (`ClosureProperties`) prove the barrier is robust under composition, reindexing, post-composition, embedding, product formation, multi-agent delegation, and nested simulation.

What is not being claimed. The theorems are unconditional mathematics; the *application* to any particular domain requires domain-specific premises (P2 for consciousness, P3 for computational self-models, `ChartableR2` biconditionals for a specific state-space model, etc.). The formalism does not prove that consciousness is non-representational, that AI cannot be conscious, or that symbol grounding is impossible. It proves that *if* the relevant state space has the boundary pattern characterized here, *then* no homeomorphism to a space with a different boundary pattern exists—and that the six most natural attempts to route around that barrier fail in formally checkable ways.

The admissibility picture. The horn closures (§7.11) together with the admissibility analysis (§7.12) leave a clean landscape. All three pillars—diagonal (Pillar 1), topological (Pillar 2), and ontological sort (Pillar 3)—contribute to the closure. In each neighboring debate—consciousness, symbol grounding, diagonal limitations, measurement chains, AI scaling, AGI self-improvement, the simulation hypothesis, and computational theories of physics and mind—the admissible solutions share a common shape: *structural change* to the topological type of the substrate, not *quantitative extension* within a fixed type. Scaling, iterating, partial modeling, finite-depth loops, cross-object readout, and decidable restriction are all formally inadmissible.

What fills the blind spot. The blind spot is not a mere absence. A functioning self-modeling system *does* produce diagonal behavior: it processes self-referential inputs and generates outputs. But no row of its self-model matches that behavior (§7.9). Therefore, the system's diagonal behavior is generated by substrate-level processes that the self-model cannot represent. The substrate is the **formally necessary complement** to any parametric self-model. Self-awareness is not

impossible—it occurs—but it occurs *with* a diagonal opacity, not without one. The blind spot is the formal signature of the substrate's irreducible contribution. Perfectly self-transparent self-awareness is the one thing that is structurally impossible.

The final picture

Every self-modeling system is provably incomplete with respect to its own diagonal. Every attempt to close the gap—scaling, iterating, delegating, restricting, embedding, changing the target topology—is formally blocked. Yet the system functions. Therefore, **something outside the self-model generates the diagonal behavior**: the substrate. This is not optional; it is a deductive consequence of the theorem and the empirical fact that self-modeling systems exist and operate. Representational incompleteness does not say self-awareness is impossible. It says self-awareness is **irreducibly substrate-dependent**: the self-model is a map, but the territory at the diagonal is charted only by the substrate itself. The gap between knower and known is not a deficiency to be engineered away. It is a proved, permanent, structural feature of any system that turns its gaze upon itself.

The formalism makes this precise and machine-checkable, so that future claims about circumventing the barrier can be evaluated against a fixed standard rather than debated informally.

Future work.

- Discharge additional witnesses (Klein bottle, projective plane) for the general obstruction class.
- A full IsManifold instance for MobiusStrip.
- A more abstract Lawvere statement for general cartesian closed categories.
- Native higher-categorical horn closures (2-category and ∞-category analogues).
- A full Turing-degree analogue of the decidable Lawvere horn (Rice's theorem / Kleene recursion in the same framework).

A Lean Module Map

Module	Role
`Basic`	`RepresentationalSystem`, iterates, `OntologicalSlot`
`Regress`	Non-termination, unbounded levels
`FixedPoints`	`EndoVsPoint`, Lawvere wall non-dissolution
`Lawvere`	Fixed-point theorem in `Type`
`LawvereCCCType`	`MonoidalClosed` packaging
`Orientability`	T_2 homeomorphism invariance
`CylinderMobius`	Fundamental domain, quotient, compact Hausdorff, boundary band
`CylinderBoundary`	Closed cylinder boundary: two disjoint connected faces
`CylinderChartable Boundary`	Cylinder C4 biconditional
`ChartableR2Neighbor`	`ChartableR2`, `ChartableR2BoundaryModel`; openness / stability
`ChartableR2Bridge`	`Homeomorph.chartableR2_iff`; `not_homeomorphic_of_chartableR2_boundary_contrast`; `ChartableR2BoundaryModel.not_homeomorphic`
`ChartableR2Concrete BoundaryModels`	Möbius/cylinder boundary models; conditional observer chain; `Quantum ObserverChainHypothesis`
`SymbolGrounding`	`SymbolSystem`; `symbolGrounding_*`
`LawvereRegressBridge`	`no_universal_parametric_unary_nat`, `no_universal_parametric_unary_bool`; `lawvere_diagonal_not_in_range`; `lawvere_fixed_point_stays_representational`

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Module	Role
NoEscapeClosure	Iterate collisions vs. injective towers; section–retraction idempotent readout
ChartableR2Models	Interior charts; cylinder interior ChartableR2
MobiusChartable Boundary	Open-cell charts; boundary ¬ChartableR2; Möbius C4 packaging
MobiusSeamChartable	Saturated seam patches (open, saturated)
MobiusSeamChart	IFT-based local model; trig coordinates
MobiusSeamTrigInject	Trig injectivity on quotient windows
MobiusSeamChartableR2	Seam charts (trig + unfold); interior-height; **M-FINAL**
MobiusCylinder BoundaryContrast	Boundary subspaces not homeomorphic
HalfLineVsLine	1D half-space $\neq$ line
HalfPlaneVsPlane	2D half-space $\neq$ plane
HalfSpaceNeighborVs Plane	Local half-space $\neq$ open plane patch
PuncturedPlaneNot SimplyConnected	$\mathbb{C} \setminus \{0\}$ not simply connected
OpenCompactChart Obstruction	Compact $\mathbb{R}^2$ patch $\neq \mathbb{R}^2$
ClosureProperties	Compositional inheritance; ComputationalSelfModel (P3); slot robustness; TopologicalSelfModelingSystem pillar bridge; exhaustive closure; simulation; computability independence
Main	Master theorem + extended bundle

B Key Theorem Index

Lean name	Statement	Module
representational_ incompleteness	**Summit:** $\forall a_0,\ s\, a_0 \neq \lambda a.\, f(s\, a\, a)$; $\neg\exists$ total s	LawvereRegress Bridge
regress_iterates_ unbounded	$\forall B, \exists L > B$, distinct iterate	Regress
lawvere_fixed_point_ Type	Universal enumerator $\Rightarrow$ every endo has fixed point	Lawvere
lawvere_wall_is_not_ dissolution	endo(represent) $\neq$ obj(fp)	FixedPoints
instT2SpaceMobius Strip	Möbius strip is T_2	CylinderMobius
isConnected_mobius StripBoundary	Boundary band is connected	CylinderMobius
closedCylinder_ boundary_eq_union	Cylinder boundary = 2 disjoint faces	CylinderBoundary
euclideanHalfSpace2_ not_homeomorphic_ euclidean2	$H^2 \not\cong_t \mathbb{R}^2$	HalfPlaneVsPlane
not_homeomorphic_of_ chartableR2_boundary_ contrast	Generic: incompatible boundaries $\Rightarrow \neg$ homeo	ChartableR2Bridge
closedCylinder_ boundaryUnion_iff_ not_chartableR2	Cylinder C4 biconditional	CylinderChartable Boundary
not_chartableR2_of_ mem_mobiusStrip Boundary	Möbius boundary $\Rightarrow$ ¬ChartableR2	MobiusChartable Bdy

continued on next page

Lean name	Statement	Module
chartableR2_mobius QuotientMk_of_interior_height	Interior height ⇒ ChartableR2	MobiusSeam ChartableR2
mobiusStripBoundary_not_homeomorphic...	Boundary subtypes not homeo	MobiusCylBdy Contrast
mobiusStrip_not_homeomorphic_closedCylinder	**M-FINAL**	MobiusSeam ChartableR2
representational_regress_master	Core master conjunction	Main
representational_regress_master_extended	Extended bundle (+ topology)	Main
QuantumObserverChainHypothesis.not_homeomorphic_mobiusStrip	Conditional: obs. chain $\not\simeq_t$ Möbius	ChartableR2 ConcreteBoundaryModels
no_universal_parametric_unary_nat	No universal $s : A \to A \to \mathbb{N}$ for all $g : A \to \mathbb{N}$	LawvereRegressBridge
IterateClosure.End.pow_iterate_dichotomy	Injective iterate powers vs. finite index bound	NoEscapeClosure
CrossObjectRepresentation.section_retraction_no_injective_tower_either_side	Both-side: ¬ injective $\mathbb{N}$-powers on source *or* host	NoEscapeClosure
lawvere_diagonal_not_in_range	Diagonal never in range of any $s : A \to A \to B$ (no surjectivity)	LawvereRegressBridge
no_universal_parametric_unary_bool	No universal $s : A \to A \to \mathrm{Bool}$ for all $g : A \to \mathrm{Bool}$	LawvereRegressBridge
Closure properties (ClosureProperties)		
ComputationalSelfModel.diagonal_blind_spot	P3 bridge: self-model has diagonal blind spot	ClosureProperties
inherited_diagonal_exclusion	Subsystem blind spot propagates to containing system	ClosureProperties
blind_spot_survives_reindexing	Reindexing via $r : C \to A$ preserves blind spot	ClosureProperties
recursive_self_improvement_blind_spot	Every link in an improvement chain has blind spot	ClosureProperties
multiagent_delegation_blind_spot	Delegated model via $\pi : A_1 \to A_2$ has blind spot	ClosureProperties
OntologicalSlot.wall_survives_injection	Wall persists under injective relabeling	ClosureProperties
blind_spot_survives_postcomposition	Post-composed model $h \circ s$ has blind spot	ClosureProperties
crossObject_both_sides_blind_spot	Embedding: A-side and C-side both have blind spot	ClosureProperties
product_self_model_blind_spot	Product of self-modeling systems has blind spot	ClosureProperties
TopologicalSelfModelingSystem.both_pillars_apply	Pillar bridge: diagonal + topological simultaneously	ClosureProperties

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Lean name	Statement	Module
combined_diagonal_and_topological_barrier	Conjunction: blind spot ∧ ¬surjective ∧ ¬homeo	ClosureProperties
pillar1_full_closure	Full Pillar 1: diagonal + no surjection + ℕ + Bool	ClosureProperties
diagonal_exclusion_is_constructive	Diagonal exclusion needs no LEM	ClosureProperties

C Theorem Dependency Graph

The following diagram shows the logical dependency structure among the principal theorems. An arrow $A \to B$ means "B uses A as a lemma or hypothesis." The three pillars are shaded.

Layer 1: Diagonal & Categorical.

- lawvere_fixed_point_Type → diagonal-not-in-range, no-universal-nat, no-universal-bool → [**P1**] representational_incompleteness
- OntologicalSlot (def) → [**P3**] lawvere_wall_is_not_dissolution → lawvere_fixed_point_stays_representational
- regress_iterates_unbounded + P3 → representational_regress_master
- pow_iterate_dichotomy (Horn 2); section–retraction closure (Horn 3)

Layer 2: Topological Obstruction.

- Homeomorph.chartableR2_iff → [**P2**] not_homeomorphic_of_chartableR2_boundary_contrast
- Cylinder C4 + Möbius C4 + boundary contrast → mobiusStrip_not_homeomorphic_closedCylinder **(M-FINAL)**
- Interior-height chartability → Möbius C4; half-space obstruction → Cylinder C4

Layer 3: Integration.

- representational_regress_master + M-FINAL → representational_regress_master_extended

Layer 4: Closure Properties.

- Summit → inherited-diagonal, reindexing, post-composition, product-model, exhaustive-closure
- OntologicalSlot → wall-survives-injection, iterate-tower-injective
- ComputationalSelfModel (P3) → diagonal-blind-spot, no-complete-self-model
- cross-object → both-sides-blind-spot, delegation-blind-spot
- TopologicalSelfModelingSystem → both-pillars-apply, combined-barrier
- pillar1/2/3-full-closure + categorical-horn-closures

Summary statistics. The Lean development contains 361 theorems and 98 lemmas across 30 source files, with zero sorry and no axioms beyond Lean 4's standard foundation (propositional extensionality, quotients, choice). The Key Theorem Index (Appendix B) lists the 35 principal results; all are verified present in the source.

D Reproduction

```
cd representational-incompleteness-lean
lake update
lake exe cache get
lake build RepresentationalIncompleteness
# Verify: 0 sorry
rg "sorry" RepresentationalIncompleteness/
```

Toolchain: leanprover/lean4:v4.29.0-rc6. Mathlib: v4.29.0-rc6.